



SCEGGS Darlinghurst

2003
Higher School Certificate
Trial Examination

Mathematics

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.

Total marks - 120

- Attempt Questions 1–10
- All questions are of equal value.

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Centre Number

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Student Number

Total marks – 120

Attempt Questions 1–10

All questions are of equal value

Answer each question on a NEW page.

Question 1 (12 marks)

Marks

- (a) Evaluate, correct to two significant figures,

2

$$\frac{195 \cdot 32}{4 \cdot 6^2 + 5 \cdot 73}$$

- (b) Solve $\frac{y}{4} - \frac{y-6}{8} = 2$.

2

- (c) Solve $x^2 + 3x > 10$

2

- (d) State the range of:

1

$$y = (x - 1)^2 + 4$$

- (e) Express $\frac{1}{\sqrt{2}} + \frac{3}{\sqrt{2}-1}$ as a single fraction with a rational denominator.

3

- (f) Simplify fully:

2

$$\log_a a^2 - \log_a \frac{1}{a}$$

Marks

Question 2 (12 marks) Start a NEW page.

A(2, 4) and B(8, 12) are the ends of a diameter of a circle.

(a) Find the co-ordinates of the centre, C, of the circle.

1

(b) Find the radius of the circle.

1

(c) State the equation of the circle.

1

(d) Hence show that D(5, 13) lies on the circle.

1

(e) Show that $AD \perp BD$.

2

(f) The perpendicular bisector of AB meets the circle at X and Y.
Find the equation of XY in general form.

2

(g) Show that the area of $\triangle XDY$ is 20m^2

2

(h) Show that $3x + 4y - 22 = 0$ is a tangent to the circle.

2

Marks

Question 3 (12 marks) Start a NEW page.

(a) A yacht sails from Robinson Island on a bearing of 240° for 120km.
It then turns and sails on a bearing of 110° until it reaches a destination
due south of its original position.

3

Calculate the distance of the yacht from Robinson Island to the nearest kilometre.

(b) Differentiate:

(i) $y = \ln(x^2 + 1)$

1

(ii) $y = \frac{e^{2x}}{\sin 3x}$

2

(c) Evaluate

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x^2 - x - 6}$$

1

(d) Solve $\sin^2 \theta - \sin \theta - 2 = 0$, $-\pi \leq \theta \leq \pi$.

3

(e) α and β are the roots of the equation $3x^2 - 6x + 2 = 0$

2

Find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$

Question 4 (12 marks) Start a NEW page.

Marks

- (a) Evaluate $\sum_{r=1}^4 r^2 - 1$ 1
- (b) Find:
- (i) $\int 3\sqrt{x} + \frac{1}{x^4} dx$ 2
- (ii) $\int_0^2 (e^{3x} + e^{-5x})^2 dx$ 3
- (c) Show that: 3
- $$\frac{2\cos^3 \theta - \cos \theta}{\sin \theta \cos^2 \theta - \sin^3 \theta} = \cot \theta$$
- (d) (i) Sketch the curve $y = 3 \sin 2x$ in the domain $0 \leq x \leq 2\pi$ showing the main features of the graph. 2
- (ii) Hence use your graph to find the number of solutions to the equation $3 \sin 2x - 1 = 0$ for $0 \leq x \leq 2\pi$. 1

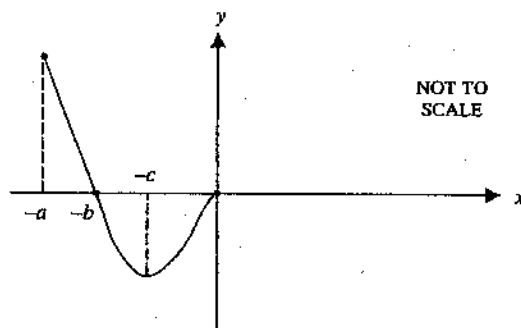
Question 5 (12 marks) Start a NEW page.

Marks

- (a) A bottle of water is placed in the common room fridge where the temperature is maintained at 0°C . The rate at which the temperature of the water falls is proportional to its temperature at that time. ($\frac{dT}{dt} = -kT$ where T is its temperature.) When the water is placed in the fridge its temperature is 40°C and after 17 minutes its temperature is 24°C .
- (i) Show that the function $T = Ce^{-kt}$ satisfies the equation $\frac{dT}{dt} = -kT$. 1
- (ii) Find the value of the constant C . 1
- (iii) Show the exact value of k is $\frac{1}{17} \ln \left(\frac{5}{3} \right)$. 2
- (iv) Find the temperature of the water after 43 minutes to the nearest degree. 1
- (b) A sheep, grazing in a paddock, is tethered to a stake by a rope 20m long. If the stake is 10m from a long fence, find the area over which the sheep can graze. 3
- (c) One set of cards contains the numbers 1, 2, 3, 4, 5 and another set contains the letters H, O, L, L, Y. One card is selected at random from each set. Find the probability of selecting:
- (i) a 4 and an H. 1
- (ii) an odd number and a vowel. 1
- (iii) a number less than 3 or an L. 2

Question 6 (12 marks) Start a NEW page.

- (a) The diagram shows the graph of a function $y = f(x)$, for $-a \leq x \leq 0$.



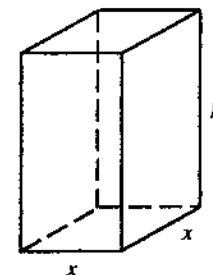
It is known that $f(x)$ is an odd function and is stationary at $(0, 0)$.

- (i) Sketch the graph $y = f(x)$, for $-a \leq x \leq a$. 1
- (ii) On a separate diagram, sketch $y = f'(x)$ 2
- (b) In a new quiz show called "Dopier than Ever", you win \$6 000 for answering the first question correctly, \$14 000 for answering the second question correctly and \$22 000 for answering the third question correctly and so on for the following questions. You finish when you answer a question incorrectly. Your total winnings for the contest is the sum of money you win on each question.
- (i) What is the prize money for the 10th question only? 2
- (ii) How many questions must you correctly answer to exceed \$1 000 000 in total winnings? 3

Question 6 continues on page 8

Question 6 (continued)

- (c) A box in the shape of a square prism has a volume of 32cm^3 and no lid. The square base has length x cm and the box is h cm high.



- (i) Show that the surface area of the box is given by: $SA = x^2 + \frac{128}{x}$. 1
- (ii) Find the dimensions of the box that has the least surface area. 3

Question 7 (12 marks) Start a NEW page.

- (a) Show that $x^3 - x^2 - x + 1 = (x+1)(x-1)^2$. 1
- (b) Consider the function, $f(x) = \frac{x-1}{x^2}$.
- (i) Prove that $f'(x) = \frac{2-x}{x^3}$. 2
- (ii) Find the co-ordinates of the stationary point on $y = f(x)$ and determine its nature. 2
- (iii) Find the co-ordinates of P, the only point where $y = f(x)$ meets the x-axis. 1
- (iv) Sketch $y = f(x)$ showing all important features. 3
- (v) Show that the equation of the tangent at P is given by the equation $y = x - 1$. 1
- (vi) Using part (a) or otherwise, find the co-ordinates of the other point where this tangent meets the curve. 2

Question 8 (12 marks) Start a NEW page.

- (a) The following is a table of values for the function $y = \frac{2}{x(x+1)}$.

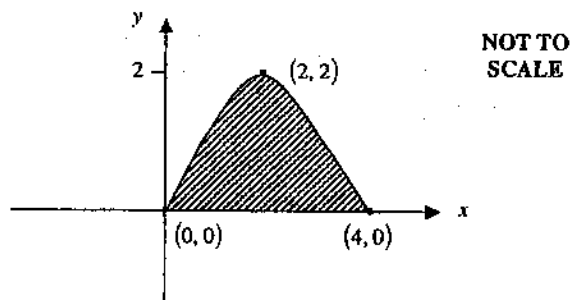
x	1	2	3	4	5
y	1	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{10}$	$\frac{1}{15}$

- (i) Using the information in the table and Simpson's rule with 5 function values, find an approximation for $\int_1^5 \frac{2}{x(x+1)} dx$ correct to 3 decimal places. 2
- (ii) It is also true that $\frac{2}{x(x+1)} = \frac{2}{x} - \frac{2}{x+1}$. Use direct integration to find the value of $\int_1^5 \frac{2}{x(x+1)} dx$ correct to 3 decimal places. 3
- (iii) Explain the difference between your answers in parts (i) and (ii). 1
- (b) A particle moves in a straight line so that its velocity v in metres per second at time t is given by $v = 4 - 2t$. At time $t = 0$ the particle is at $x = 1$.
- (i) Find the displacement x of the particle as a function of t . 2
- (ii) When is the particle at rest and what is its acceleration at that time? 2
- (iii) Find the distance the particle travels in the first 4 seconds. 2

Question 9 (12 marks) Start a NEW page.

- (a) Can there be a geometric series with a limiting sum of $\frac{2}{3}$ and a first term of 4? 2
Justify your answer with appropriate calculations.

- (b) The producers of Play School are replacing the Arched Window. It will still have a base length of 4m and a height of 2m as shown in the diagram below.



The new arch is to be either an arc of a parabola or a half-cycle of a sine curve.

- (i) If the arch is the arc of a parabola, the equation of the curve is of the form: 1

$$f(x) = ax(4-x)$$

Show that the value of a is $\frac{1}{2}$.

- (ii) If the arch is a sine curve, the equation of the curve is of the form, 1

$$g(x) = A \sin \frac{\pi x}{4}$$

Find the value of A .

- (iii) Calculate the area for each window design and hence decide which would be cheaper to build. 4

- (c) A rectangular lawn is 60 metres long and 30 metres wide. A pigeon wanders randomly around the lawn. Find the probability that the pigeon is:

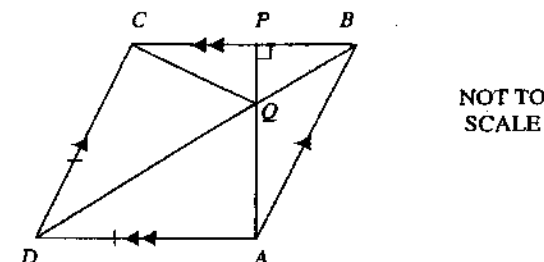
- (i) more than 10 metres from the edge of the lawn. 2

- (ii) not more than 10 metres from a corner of the lawn. 2

Question 10 (12 marks) Start a NEW page.

- (a) Consider the function $y = xe^{-2x}$. 3
Prove that $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$

- (b) In the rhombus $ABCD$, AP is constructed perpendicular to BC and intersects the diagonal BD at Q .



- (i) State why $\angle ADB = \angle CDB$. 1

- (ii) Prove that $\triangle AQD \cong \triangle CQD$. 2

- (iii) Hence find $\angle QCD$. 1

- (c) After several mornings of horrendous traffic, Chris decides to move closer to work. She takes out a loan for \$500,000 at an interest rate of 12% p.a. compounded monthly for 20 years.

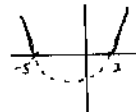
- (i) Show that the amount she owes on the loan after n months, A_n , is given by the expression: 3

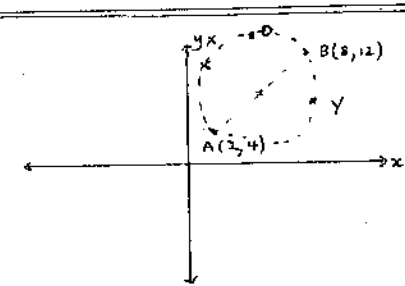
$$A_n = 100M - 1.01^n (100M - 500,000)$$

where M is the size of her monthly repayments.

- (ii) Her repayments are fixed at \$5505 per month. In which year does Chris still owe \$250,000? 2

END OF PAPER

YR 12 : 2 UNIT		
MATHEMATICS TRIAL 2003		
SOLUTIONS + MARKING SCALE		
QUESTION 1:	MF	COMMENTS
(a) 7.26366...	✓	Rounding off to 2 s.f. <u>not</u> well done. Does not mean 2 decimal places.
$\hat{=} 7.3$ correct to 2 s.f.	✓	
(b) $\frac{y}{4} - \frac{y-6}{8} = 2$		
$\therefore 2y - y + 6 = 16$	✓	this sign was a problem.
$\therefore y = 10$	✓	
(c) $x^2 + 3x - 10 > 0$		
$(x+5)(x-2) > 0$	✓	You <u>must</u> factorise, <u>sketch</u> and solve the quadratic inequality from your sketch. It is <u>incorrect</u> to solve this way $x+5 > 0$ $x-2 > 0$ $x > -5$ $x > 2$ Please <u>don't</u> do it.
$\therefore x < -5$ and $x > 2$	✓	
		
(d) $y = (x-1)^2 + 4$		
Range: $y \geq 4$	✓	Students who sketched the parabola were the most successful in finding the range.
(e) $\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	✓	
$\frac{3}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} = \frac{3\sqrt{2}+3}{1}$	✓	
$\therefore \frac{1}{\sqrt{2}} + \frac{3}{\sqrt{2}-1} = \frac{\sqrt{2}}{2} + \frac{1(3\sqrt{2}+3)}{2}$		
$= \frac{7\sqrt{2}+6}{2}$	✓	
(f) $\log_a a^2 - \log_a a^{-1} = 2 - -1$	✓	1 mark for correct use of a log rule.
$= 3$	✓	

QUESTION 2:	KB	COMMENTS
		
(a) Centre: $(\frac{2+8}{2}, \frac{4+12}{2}) = (5, 8)$	✓	Good
(b) $d = \sqrt{(12-8)^2 + (8-5)^2}$		Good
$= \sqrt{16+9}$		
$= 5$	✓	
(c) Equation:		Good
$(x-5)^2 + (y-8)^2 = 5^2$	✓	
(d) When $x = 5$:		Good
$(5-5)^2 + (y-8)^2 = 5^2$		
$0 + (y-8)^2 = 25$		
$y-8 = \pm 5$	✓	
$\therefore y = 13$ or 3		
$\therefore (5, 13)$ does lie on the circle		
(e) $m_{AD} = \frac{13-4}{5-2} = \frac{9}{3} = 3$		Both gradients correct for 1 mark.
$m_{BD} = \frac{13-12}{5-8} = -\frac{1}{3}$	✓	
$\therefore m_{AD} \times m_{BD} = 3 \times -\frac{1}{3} = -1$		
$\therefore AD \perp BD$	✓	
(f) $m_{AB} = \frac{12-4}{8-2} = \frac{8}{6} = \frac{4}{3}$		not well done. A diagram would assist in the next sections <u>must</u> be in general form.
$\therefore \text{Grad } \perp AB : -\frac{3}{4}$	✓	
$y-8 = -\frac{3}{4}(x-5)$		
$3x+4y-47=0$	✓	

Question 2 (cont.)

(g) $XY = 10$ units (diameter of circle). ✓

⊥ height is dist of D from XY:

$$d = \frac{|3 \times 5 + 4 \times 13 - 47|}{\sqrt{9+16}}$$

$$= \frac{20}{5} = 4$$

$$\therefore \text{Area} = \frac{1}{2} \times 10 \times 4 = 20 \text{ u}^2$$

(h) ⊥ distance of line from centre:

Centre: (5, 8)

$$d = \frac{|3 \times 5 + 4 \times 8 - 22|}{\sqrt{3^2 + 4^2}}$$

$$= \frac{25}{5}$$

$$= 5$$

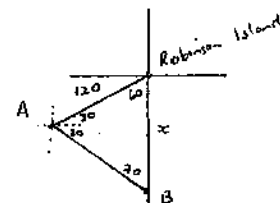
Since the line is 5 units from the centre of the circle of radius 5, it is a tangent to the circle. ✓

QUESTION 3:

Com 1
Calc 3 CB

COMMENTS

(a)



$$\angle RBA = 70^\circ$$

$$\frac{x}{\sin 50} = \frac{120}{\sin 70}$$

$$\therefore x = 97.824 \dots$$

$$\approx 98 \text{ km}$$

(b) (i) $y = \ln(x^2 + 1)$

$$\frac{dy}{dx} = \frac{2x}{x^2 + 1}$$

$$(ii) y = \frac{e^{2x}}{\sin 3x}$$

$$u = e^{2x} \quad v = \sin 3x$$

$$u' = 2e^{2x} \quad v' = 3\cos 3x$$

$$\frac{dy}{dx} = \frac{2e^{2x} \sin 3x - 3e^{2x} \cos 3x}{(\sin 3x)^2}$$

$$= \frac{e^{2x} (2 \sin 3x - 3 \cos 3x)}{(\sin 3x)^2}$$

$$(c) \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{(x-3)(x+2)} = \frac{6}{5}$$

$$(d) \sin^2 \theta - \sin \theta - 2 = 0$$

$$(\sin \theta - 2)(\sin \theta + 1) = 0$$

$$\therefore \sin \theta = 2 \quad \text{or} \quad \sin \theta = -1$$

no solution

$$\theta = -\frac{\pi}{2}$$

$$(e) \alpha + \beta = 2$$

$$\alpha \beta = \frac{2}{3}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha \beta}$$

$$= 2 \div \frac{2}{3}$$

$$= 3$$

Com 1 - diagram only

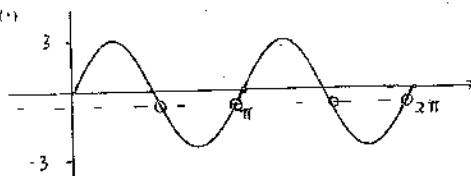
must round to nearest km.

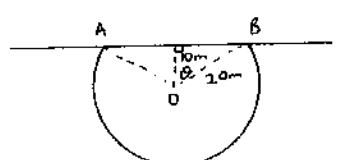
(b) Calc 3

Several students failed to put v^2 in the denominator (further simplification is not necessary) ignore subsequent errors

(must not just ignore no solⁿ case).

Well done.

QUESTION 4:	Com 2 Calc 5 Reas 3	HG	COMMENTS
(a) $\sum_{r=1}^4 r^2 - 1 = 0 + 3 + 8 + 15 = 26$ ✓			• Some students forget to <u>add</u> !
(b) (i) $\int 3\sqrt{x} + x^{-4} dx$ $= \frac{3x^{3/2}}{3/2} + \frac{x^{-3}}{-3} + C$ $= 2\sqrt{x^3} - \frac{1}{3x^3} + C$	✓	✓	• Learn index rules carefully. further simplification <u>not</u> required + C = 2nd mark.
(ii) $\int_0^2 (e^{5x} + e^{-5x})^2 dx$ $= \int_0^2 e^{10x} + 2 + e^{-10x} dx$ $= \left[\frac{1}{10} e^{10x} + 2x + \frac{1}{10} e^{-10x} \right]_0^2$ $= \left(\frac{1}{10} e^{20} + 4 - \frac{1}{10} e^{-20} \right) - \left(\frac{1}{10} + 0 - \frac{1}{10} \right)$ $= \frac{1}{10} e^{20} + 4 - \frac{1}{10} e^{-20}$ ✓	✓	✓	(b) Calc 5 (i) Poorly done • Expansion poor • Many integration errors eg $\int e^{10x} dx \neq \frac{1}{10} e^{10x} + C$ ← must show evidence of substituting $x=0$!
(c) LHS: $\frac{\cos \theta (2 \cos^2 \theta - 1)}{\sin \theta (\cos^2 \theta - \sin^2 \theta)}$ $= \frac{\cos \theta (2 \cos^2 \theta - 1)}{\sin \theta (\cos^2 \theta - (1 - \cos^2 \theta))}$ $= \frac{\cos \theta (2 \cos^2 \theta - 1)}{\sin \theta (2 \cos^2 \theta - 1)}$ $= \cot \theta$ $= RHS$	✓	✓	c) Reas 3 • many students gave up after factoring! substitute by expression. (Ext student may use $\cos 2\theta$) simplify.
(d) (i) 	✓	✓	✓ period ✓ amplitude (i) Com 2 • Well done!
(ii) 4 solutions to the equation.	✓	✓	(or c/s).

QUESTION 5:	Calc 5 Reas 5	MF	COMMENTS
(a) (i) $T = Ce^{-kt}$ $\frac{dT}{dt} = -k \times Ce^{-kt}$ $= -kT$	✓		(a) Calc 5
(ii) when $t=0$, $T=40$ $\therefore 40 = Ce^0$ $\therefore C=40$	✓		
(iii) when $t=17$, $T=24$ $\therefore 24 = 40e^{-k \times 17}$ $\frac{24}{40} = e^{-17k}$ $\therefore \ln\left(\frac{3}{5}\right) = -17k$ $\therefore k = -\frac{1}{17} \ln\left(\frac{3}{5}\right)$ $= \frac{1}{17} \ln\left(\frac{5}{3}\right) \approx 0.03$ ✓	✓		First mark given for knowing to take logs both sides and using a log-law. The log-law used here should really be shown in working. No students did this so the mark was awarded for having the required answer.
(iv) $T = 40 \times e^{-\frac{1}{17} \ln\left(\frac{5}{3}\right) \times 43}$ $= 10.982 \dots$ $\approx 11^\circ$	✓		
(b) 			(b) Reas 3
$\cos \theta = \frac{1}{2}$ $\therefore \theta = \frac{\pi}{3}$ $\therefore \angle AOB = \frac{2\pi}{3}$ ✓	✓		<u>Note Very important</u> Angle <u>must</u> be in radians.
Area minor seg: $\frac{1}{2} \times 20^2 \times \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3}\right)$ $= 200 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right) \approx 245.7$ ✓	✓		Required Area = Circle - minor Segment = Circle - (sector - triangle)
Area of major seg: $\pi \times 20^2 - 200 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$ $= \frac{800\pi}{3} + 100\sqrt{3}$ ≈ 1010.96 ✓	✓		

Question 5 (cont.)

COMMENTS

$$(c) (i) P(4, H) = \frac{1}{5} \times \frac{1}{5} = \frac{1}{25} \quad \checkmark$$

$$(ii) P(\text{odd, vowel}) = \frac{3}{5} \times \frac{1}{5} = \frac{3}{25} \quad \checkmark$$

$$(iii) P(L3 \text{ or } L) = \frac{2}{5} + \frac{3}{5} \times \frac{2}{5} \\ = \frac{16}{25} \quad \checkmark$$

Some students added the fractions. You must multiply the successive events probabilities together.

(iii) Ans $\frac{1}{2}$

These events in (iii) are not mutually exclusive. They have something in common.

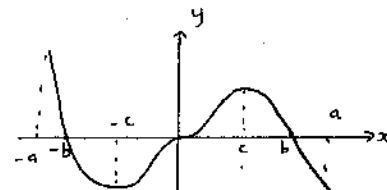
$$P(L3 \text{ or } L) = P(L3) + P(L) - P(L3 \text{ and } L) \\ = \frac{2}{5} + \frac{2}{5} - \frac{2}{5} \times \frac{2}{5} \\ = \frac{10}{25} + \frac{10}{25} - \frac{4}{25} \\ = \frac{16}{25}$$

QUESTION 6

Com 13
Calc 14
Res 13 MF

COMMENTS

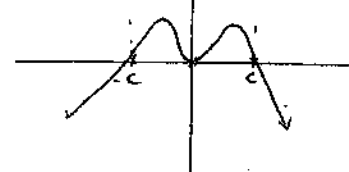
(a) (i)



(a) Com 13

✓✓✓ odd fn + correct derivative showing horiz. P.O.I.

(ii)



✓ at $x=2x$
✓ + f at $(0,0)$

✓✓ odd fn + correct derivative with no indication of horiz. P.O.I.

✓ max. for even fn. + correct derivative.

(b)

(i) 6000, 14000, 22000, ...

$$a = 6000 \quad \checkmark \\ d = 8000 \quad \checkmark \\ n = 10 \quad \checkmark \\ T_{10} = 6000 + 9 \times 8000 \\ = 78000 \quad \checkmark$$

$$(ii) S_n = \frac{n}{2} (12000 + (n-1) 8000) \\ = 6000n + 4000n^2 - 4000n \\ = 4000n^2 + 2000n \quad \checkmark$$

$$\text{But } S_n \geq 1000000$$

$$4000n^2 + 2000n - 1000000 \geq 0$$

$$2n^2 + n - 500 \geq 0 \quad \checkmark$$

$$\therefore n \leq -16.1 \quad \text{or} \quad n \geq 15.6$$

Must answer 16 questions correctly to exceed one million. ✓

(ii) Res 13

Show clear formulae, setting out + working.

Question 6 (cont.)

COMMENTS

(a) (i) Volume = 32

$$x^2 h = 32$$

$$\therefore h = \frac{32}{x^2}$$

$$S.A = x^2 + 4xh$$

$$= x^2 + 4x \cdot \frac{32}{x^2}$$

$$= x^2 + \frac{128}{x}$$

(ii) $S = x^2 + 128x^{-1}$

$$\frac{dS}{dx} = 2x - 128x^{-2}$$

$$\text{min SA} \Rightarrow \frac{dS}{dx} = 0$$

$$2x - \frac{128}{x^2} = 0$$

$$2x = \frac{128}{x^2}$$

$$\therefore x^3 = 64$$

$$\therefore x = 4$$

Dimensions are $4 \times 4 \times 2$ ✓

(c) Calc 4

Substitution of h must be clearly shown.

Students found it difficult to solve this sort of equation.

must give both x & h values.

QUESTION 7:

Com 3
Reas 2 HG

COMMENTS

(a) RHS: $(x+1)(x-1)^2$

$$= (x^2-1)(x-1)$$

$$= x^3 - x^2 - x + 1$$

$$= \text{LHS}$$

{ or by factoring LHS }

(b) (i) $f(x) = \frac{x-1}{x^2}$

$$f'(x) = \frac{x^2 \cdot 1 - 2x \cdot (x-1)}{x^4}$$

$$= \frac{x^2 - 2x^2 + 2x}{x^4}$$

$$= \frac{2x - x^2}{x^4}$$

$$= \frac{2-x}{x^3}$$

(ii) Set $p_b \Rightarrow f'(x) = 0$

$$\frac{2-x}{x^3} = 0$$

$$2-x = 0$$

$$\therefore x = 2$$

x	2 ⁻	2	2 ⁺
$f'(x)$	+ve	0	-ve

$\therefore (2, \frac{1}{4})$ is a max t.p. ✓

(iii) $f(x) = 0$

$$\frac{x-1}{x^2} = 0$$

$$\therefore x-1 = 0$$

$$\therefore x = 1$$

$$\therefore (1, 0)$$

Many students tried to fudge it!

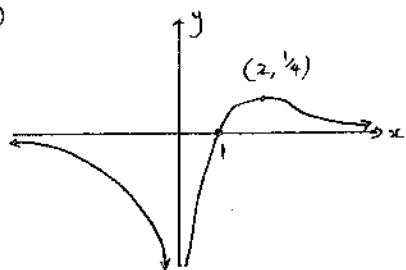
(i), (ii), (iii) well done by most students.

(or by 2nd derivative)
 $f''(x) = \frac{2x-6}{x^4}$

QUESTION 7 (cont.)

COMMENTS

(iv)



- ✓ - max
- ✓ - asymptote $x=0$
- ✓ $\lim_{x \rightarrow \pm \infty} f(x) = 0$

Com 3

Many students didn't sketch the curve for $x < 0$.

$$(v) f'(x) = \frac{2-x}{x^3}$$

$$\text{At } P(1,0), f'(x) = 1$$

$$\therefore y - 0 = 1(x - 1)$$

$$\therefore y = x - 1$$

✓

(vi) Tangent meets curve:

$$\begin{cases} y = x - 1 \\ y = \frac{x-1}{x^2} \end{cases}$$

$$\therefore x - 1 = \frac{x-1}{x^2}$$

$$\therefore x^3 - x^2 = x - 1$$

$$x^3 - x^2 - x + 1 = 0$$

✓

using (a):

$$(x+1)(x-1)^2 = 0$$

$$\therefore x = \pm 1$$

The tangent also meets the curve at $(-1, -2)$

✓

Reas 1 (vi) Poorly done.
many students did not use the hint to use part (a).

QUESTION 8

Calc 6
Com 1 CB

COMMENTS

$$(a) \int_1^5 \frac{2}{x(x+1)} dx$$

$$= \frac{1}{3} \left(1 + 4\left(\frac{1}{3} + \frac{1}{10}\right) + 2\left(\frac{1}{6} + \frac{1}{15}\right) \right)$$

$$= \frac{1}{3} \times 3^{2/5}$$

$$= 1.044$$

✓

$$(ii) \int_1^5 \frac{2}{x(x+1)} dx$$

$$= \int_1^5 \frac{2}{x} - \frac{2}{x+1} dx$$

$$= [2 \ln x - 2 \ln(x+1)]_1^5$$

✓

$$= 2 \ln 5 - 2 \ln 6 - 2 \ln 1 + 2 \ln 2$$

✓

$$= 2 \ln \frac{5}{3}$$

$$= 1.022$$

✓

(iii) Simpson's rule is an approximation for the integral (using parabolic arcs) whereas (ii) calculated the exact value of the integral.

Several students chose $n=5$ rather than $n=4$ and several confused the formula.

Generally well done by those who used the appropriate substitution.

Com 1
Mostly well understood.

Question 8 (cont.)

COMMENTS

(b) $v = 4 - 2t$

(i) $x = 4t - t^2 + C$ ✓

when $t = 0$, $x = 1$

$\therefore 1 = 0 - 0 + C$

$\therefore C = 1$

$\therefore x = 4t - t^2 + 1$ ✓

(ii) Rest $\Rightarrow v = 0$

$0 = 4 - 2t$

$\therefore t = 2$ ✓

$a = -2$ at all time, t . ✓

(iii) At $t = 0$, $x = 1$

$t = 2$, $x = 5$ ✓

$t = 4$, $x = 1$

$\therefore$ Distance travelled $= 4 + 4$

$= 8$ metres ✓

(b) Calc 6

Only a few students were able to manage this question.

QUESTION 9

Reas 8 CB

COMMENTS

(a) If $a = 4$ and $S_{\infty} = \frac{2}{3}$

then $\frac{2}{3} = \frac{4}{1-r}$

$2 - 2r = 12$

$\therefore r = -5$ ✓

But $|r| < 1$ for S_{∞} to exist.

$\therefore$ No series exists. ✓

(b) (i) The parabola must pass through $(2, 2)$

$\therefore 2 = a \times 2(4 - 2)$

$2 = 4a$ ✓

$\therefore a = \frac{1}{2}$

(ii) A is the amplitude

$\therefore A = 2$ ✓

(iii) Parabola: $A = \int_0^4 \frac{1}{2}x(4-x) dx$

$= \int_0^4 2x - \frac{1}{2}x^2 dx$

$= \left[x^2 - \frac{1}{6}x^3 \right]_0^4$ ✓

$= 16 - \frac{32}{3}$

$= \frac{16}{3} \text{ units}^2$ ✓

Sine wave $A = \int_0^4 2 \sin \frac{\pi x}{4} dx$

$= \left[-\frac{8}{\pi} \cos \frac{\pi x}{4} \right]_0^4$ ✓

$= -\frac{8}{\pi} \cos \pi + \frac{8}{\pi} \cos 0$

$= \frac{16}{\pi} \text{ units}^2$ ✓

Sine wave is smaller & hence cheaper to build

Students did not generally know the condition for a limiting sum with several incorrectly stating that $|r| \leq 1$.

Other students did not interpret/read the question and substituted $r = \frac{2}{3}$ instead of $S_{\infty} = \frac{2}{3}$

Several students failed to recognize that the problem could be solved by mere substitution of a point into the equation

(iii) Reas 4

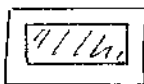
Product rule does not apply to integration!

Few students were able to handle this integration

QUESTION 9 (cont.)

COMMENTS

(c) (i)

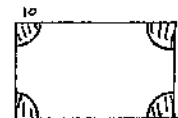


$$\begin{aligned} \text{Lawn} &= 1800 \quad \checkmark \\ \text{Area} &= 40 \times 10 \\ &= 400 \\ \therefore P &= \frac{400}{1800} \\ &= \frac{2}{9} \quad \checkmark \end{aligned}$$

(c) Reas 4

Well done.

(ii)



$$\begin{aligned} \text{Lawn} &= 1800 \\ \text{Area} &= \pi \cdot 10^2 \\ &= 100\pi \quad \checkmark \end{aligned}$$

$$\begin{aligned} \therefore P &= \frac{100\pi}{1800} \\ &= \frac{\pi}{18} \quad \checkmark \end{aligned}$$

QUESTION 10

Calc 3
Reas 9 HG

COMMENTS

(a) $y = x e^{-2x}$

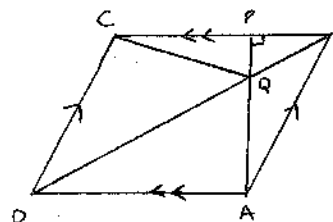
$$\begin{aligned} \frac{dy}{dx} &= e^{-2x} \cdot 1 + x \cdot (-2e^{-2x}) \\ &= e^{-2x} - 2xe^{-2x} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= -2e^{-2x} - 2(e^{-2x} - 2xe^{-2x}) \\ &= -4e^{-2x} + 4xe^{-2x} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \therefore \frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y \\ &= -4e^{-2x} + 4xe^{-2x} + 4e^{-2x} - 8xe^{-2x} + 4xe^{-2x} \\ &= 0 \end{aligned}$$

(a) Calc 3 . Many students forgot to do the product rule!

(b)



(b) Reas 4

(i) $\angle APB = \angle CDB$ (diagonals of a rhombus bisect the angles they pass through).

(ii) $\angle ADB = \angle CDB$ (as above)

DQ is common

CD = AD (given)

$\therefore \triangle AQB \cong \triangle CQB$ (SAS)

(iii) $\angle QAD = 90^\circ$ (alt $\angle$ as $CB \parallel AD$)

$\therefore \angle QCD = 90^\circ$ (corr. $\angle$ in $\triangle$ are $\equiv$)

Reasons - poor.

QUESTION 10 (cont.)	COMMENTS
(c) (i) $A_1 = 500000 (1.01) - M$ $A_2 = A_1 \times 1.01 - M$ $= 500000 (1.01)^2 - M(1.01) - M$ $A_3 = A_2 \times 1.01 - M$ $= 500000 (1.01)^3 - M(1.01)^2 - M(1.01) - M$ $\vdots$ $\therefore A_n = 500000 (1.01)^n - M [1.01^{n-1} + 1.01^{n-2} + \dots + 1]$ $\uparrow$ GP $a = 1$ $r = 1.01$ $n = n$ $\therefore S_n = 1 \frac{(1.01^n - 1)}{1.01 - 1}$ $= \frac{1.01^n - 1}{0.01}$ $\therefore A_n = 500000 (1.01)^n - M \times \left(\frac{1.01^n - 1}{0.01} \right)$ $= 500000 (1.01)^n - 100M (1.01^n - 1)$ $= 100M - 1.01^n (100M - 500000)$ (ii) $250000 = 100 \times 5505 - 1.01^n (100 \times 5505 - 500000)$ $250000 = 550500 - 1.01^n (50500)$ $\therefore 1.01^n = \frac{550500 - 250000}{50500}$ $= 5.95$ $\therefore n = \frac{\ln 5.95}{\ln 1.01}$ $= 179.23 \text{ months}$ $\therefore$ at the 15th year.	2.5 / 5 . Many attempt at fudging - not very successful. Many students still got 2 marks in (ii) despite being stuck in (i).